



Quantum group decision making based on regret theory and grey relational degree considering psychological preference

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ABSTRACT

To address the mutual interference among decision-makers (DMs), this paper proposes a quantum group decision making model that incorporates psychological preferences to characterize the interference mechanism in the process of group opinion aggregation. Firstly, the regret utility values of DMs for each alternative are calculated, and the DM's weight function is constructed based on the distance of each DM's regret utility value from the group minimum. Secondly, a psychological preference utility function is proposed to characterize the discrepancy between individual and group evaluations, and an optimization model is constructed accordingly to determine the optimal attributes' weights. Meanwhile, a quantum-like Bayesian network (QLBN) is constructed to handle the interference effect among DMs, the interference value based on the inverse grey relational degree is proposed. Finally, the alternatives are ranked according to the quantum probabilities. The validity of the model is demonstrated through a logistics equipment selection case analysis. Sensitivity analysis and comparative analysis demonstrate the robustness and adaptability of the model. The results show that the proposed model can effectively measure the interference effect among DMs while taking into account their preferences in an uncertain decision-making context.

1. Introduction

Multi-attribute group decision-making (MAGDM), as a pivotal branch of decision science, aims to support DMs in reasonably evaluating multiple attributes across different alternatives in order to make an informed decision (Ji et al., 2024; Zhu et al., 2025). Different from the traditional single DM problem, MAGDM emphasizes how to synthesize their preferences, knowledge and experience with the participation of multiple DMs (Jiang et al., 2023; Tang et al., 2025; Xing et al., 2025). In practical applications, MAGDM methods have found extensive utility across resource allocation (Fan et al., 2024), portfolio (Wang, Pan, et al., 2024), supplier selection (Liu et al., 2024) and other fields, demonstrating particular value when handling complex decision-making problems under uncertainty (Cao et al., 2024; Ji, Wu, Chiclana, Sun & Herrera-Viedma, 2024; Liao et al., 2022).

In cases where information is ambiguous, traditional decision-making models based on precise numerical values have certain limitations in expressing the internal judgement and subjective preferences of decision-makers, as well as in matching their cognitive characteristics (Dong & Wan, 2024). Therefore, the introduction of MAGDM

methods capable of handling fuzzy information is usually more applicable and interpretable in such scenarios. Equipment selection of intelligent logistics systems is a typical application scenario among them. To achieve precise distribution from the warehouse to the assembly line, enterprises need to select the optimal alternative from multiple automated devices in a fuzzy and dynamic environment. This decision involves multiple conflicting attributes such as distribution efficiency, stability, and cost, and often faces uncertain information such as flexibility evaluation and capacity estimation. Therefore, how to accurately represent the DMs' assessment of attributes has become the key to optimizing resource allocation and improving logistics efficiency. To date, fuzzy numbers (Ma et al., 2025; Xu et al., 2024) have been extensively employed for characterizing DMs' information in these problems, such as triangular fuzzy numbers (Wang, 2021), trapezoidal fuzzy numbers (Wan et al., 2021), probabilistic linguistic numbers (Wang et al., 2025), intuitionistic fuzzy numbers (Kumar & Kumar, 2024), hesitant fuzzy numbers (Ranjbar et al., 2022), etc. However, these methods all have limitations when dealing with complex decisions such as the selection of logistics equipment, which require the

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consideration of both quantitative indicators and qualitative descriptions: triangular fuzzy numbers, although structurally simple and easy to understand, have a relatively limited range of expressed fuzziness; trapezoidal fuzzy numbers, by introducing two vertices to describe fuzzy information more precisely, still lack the adaptability to dynamic changes; probabilistic linguistic numbers are complex in calculation and reasoning, and when facing dynamic changing environments, they update and adjust at a slower speed; intuitionistic fuzzy numbers usually do not clearly distinguish "the uncertainty of the evaluation itself" from "the reliability of the evaluation information"; hesitant fuzzy numbers lack a description of confidence or reliability for each possible value. Compared with the previous three types of fuzzy numbers, linguistic Z-numbers (Li, 2024) combines the characteristics of fuzzy numbers and fuzzy sets, through precise language description and probabilistic expression combination, can better reflect the preference differences and uncertain information of DMs in dynamic changing and complex backgrounds; compared with intuitionistic fuzzy numbers, linguistic Z-numbers can explicitly distinguish the cognitive uncertainty of DMs from the credibility of evaluation information by introducing an independent reliability measure; compared with hesitant fuzzy numbers, linguistic Z-numbers assign a clear reliability degree to each language evaluation, thereby quantifying the confidence level of each possibility while representing multiple possibilities. Therefore, linguistic Z-numbers, as a flexible tool for representing fuzzy information, can effectively overcome the limitations of the above fuzzy numbers and is suitable for representing the evaluation information of DMs in complex decision-making backgrounds (Xu, Yan, Pérez, & Cabrerizo, 2025).

The weight allocation among DMs exerts substantial impacts on the outcomes of MAGDM processes. Traditional methods for calculating DMs' weights usually carry out simple weighting or direct subjective allocation based on the professional background and experience of DMs or some objective criteria (She et al., 2021a). In contrast, Vijayakumar R. (R. et al., 2024) discussed a weight allocation method based on clustering and cost optimization in his study, aiming to comprehensively consider the similarity of DMs' opinions and adjust costs in order to allocate weights more effectively. Jiang et al. (2024) proposed a weight allocation method combining cloud model and Shapley function, aiming to comprehensively consider the correlation and uncertainty among DMs, so as to realize the weight allocation of DMs. However, these methods tend to ignore the psychological preferences and individual differences of DMs, and it is difficult to fully reflect DMs' authentic preferences and their emotional factors. For example, they may experience psychological fluctuations when confronted with conflicting opinions, which can alter their preferences for certain alternatives (Liu, Xu, & Wu, 2023). Regret theory (Shen et al., 2024; Zhang et al., 2023), as a behavioural decision theory, emphasizes the influence of emotional factors on decision making. The theory proposes that DMs tend to regret or regret certain decision results when faced with multiple choices. Therefore, by including DMs' regret into their weight calculation, their real decision preferences can be reflected more accurately.

The determination of attributes' weights constitutes a fundamental step in the MAGDM procedure. The weights of attributes reflect the relative importance of each attribute in the evaluation system of DMs. Thus, the accuracy of attribute weight allocation significantly impacts the soundness and efficacy of final decisions (Wan et al., 2024; Wu et al., 2022). The assignment of attribute weights dually influences both the DMs' preference ranking of alternatives and the inter-attribute relative importance relationships (Zhu et al., 2025). Traditional attribute's weights calculation methods usually rely on expert evaluation (Yang et al., 2023), entropy weight method (Zhao et al., 2024) or AHP (Meşe Erdoğan et al., 2024). Within intricate and ambiguous decision environments, the decision preferences of DMs are not only influenced by the external environment, but also disturbed by internal factors such as personal emotions and cognitive biases (Van Dijk & De Dreu, 2021). For example, Zhao et al. (2023) established the SDVW-Choquet

integral to integrate non-additive preferences. Wang et al. (2022) constructed a new regret-joy function to simulate DMs' psychological preferences. However, these methods failed to integrate subjective and objective factors organically and could not fully consider the psychological preferences, emotional responses and cognitive differences of DMs.

Quantum decision theory is a decision theory based on the principles of quantum mechanics, which aims to solve the irrationality and uncertainty problems that traditional decision models are unable to handle (Xu, Yan, & Li, 2025). Different from classical decision theory, quantum decision theory introduces concepts such as superposition, interference and measurement of quantum states, which can better explain DMs' choice behaviour under uncertain and fuzzy information environment (Liu & Zhu, 2022; Xiao et al., 2025; Yan, Xu, et al., 2024; Zheng & Liu, 2025). The fundamental value of quantum decision theory resides in its ability to model interference effects between DMs. In collective decision-making, DMs are not only affected by the external environment and task requirements, but also by the opinions and preferences of other DMs, which will affect the final decision of each DM and further affect the accuracy of group decision making results (Y. Liu et al., 2023; Tan et al., 2024; Yan, Zeng, et al., 2024). For instance, Wan et al. (2026) proposed a consensus-reaching process for quality function deployment based on dual similarity, using quantum probability theory to determine customer weights. Meanwhile, Yan, Dong, et al. (2025) addressed the issue of heterogeneous multi-criteria group decision-making, and constructed a new model based on quantum probability theory to derive the weights of DMs and the ranking of alternatives. In MAGDM, the interference effect is usually captured by constructing QLBN, and the interference angle can quantitatively reflect the magnitude and direction of mutual interference among DMs. Therefore, accurate measurement of the interference angle is very important for quantum decision making process. At present, the measurement methods of interference effect mainly include similar heuristic algorithm (Liu & Zhu, 2022; Zheng & Liu, 2025) and entropy method (Yan, Xu, et al., 2025). The similar heuristic algorithm calculates the cosine of the angle between two DMs' preference vectors and maps it to the difference of phase angles to measure the interference angle. However, when the preference distribution is uneven, the deviation generated by its measurement is large, and it is unable to deal with complex distribution. Deng entropy measures the interference angle by quantifying the uncertainty and difference of the preference distribution of DMs, but its computational complexity is high and it is easy to cause instability of the interference angle measure due to noise or outliers in the face of small sample data, thereby reducing the robustness of the results.

The main motivations of this paper are as follows:

Motivation 1: Traditional decision-making models usually assume that DMs are independent and rational during the decision-making process, ignoring the possible mutual influence and interference effects among DMs in the actual group decision-making process. Such interference effects may lead to deviations or irrationality in the decision results.

Motivation 2: Existing methods for measuring interference angles have high data requirements, making them difficult to adapt to decision-making scenarios with high data uncertainty and complex distributions. Some methods require typical probability distributions to ensure accuracy, but in actual decision-making, data is often sparse and has complex distributions, making it difficult to meet these conditions. Some methods have limited capabilities in handling nonlinear relationships and cannot effectively capture the complex correlations between vectors, resulting in large estimation deviations of the interference angle.

Motivation 3: Traditional methods for determining attribute's weights usually rely solely on the subjective judgement of DMs or assign weights based on objective data, which is difficult to achieve effective integration of subjective and objective information. At the same time,

Table 1
List of Symbols.

Notation	Explanation	Notation	Explanations
L	Set of linguistic terms, $L = \{l_0, l_1, \dots, l_{2g}\}$, where g is a positive integer.	$\bar{z}^-$	The crisp Z-numbers corresponding to the worst linguistic Z-numbers $z^- = (A_0, B_0)$.
ζ_c	The DM's preferences for linguistic term l_c , $c = 0, 1, \dots, 2g$.	α, σ	The weights of the two components of the linguistic Z-numbers.
$H_o(l_c)$	Linguistic scale function, $o = 1, 2$.	ψ_T, ψ_T^*	The weights of the two components of the linguistic Z-numbers.
A'	Alternative set.	μ_{ij}^k	regarding the evaluation of alternative $a_i (i = 1, 2, \dots, m)$ on attribute $c_j (j = 1, 2, \dots, n)$.
Cn	Attribute set.	R^k	The regret matrix.
D	DM set.	δ, p	Regret aversion coefficient, risk aversion coefficient.
W^k	The set of attribute's weights.	q_i^k	The regret utility value of DM d_k for alternative a_i .
Λ	The set of DMs' weights.	q_i^*	The group minimum regret utility value of alternative a_i .
A	Linguistic term, $A \in L_1$.	Δq_i^k	The difference between DM d_k and the group's minimum regret utility.
B	Linguistic term, $B \in L_2$.	$h\left(\frac{\bar{z}^k}{z_{ij}}\right)$	Preference utility function.
$\bar{z}_{ij}^k$	Linguistic Z-numbers.	$\bar{z}_{ij}^0$	The reference value of the group's evaluation of alternative a_i on attribute c_j .
Z^k	Initial evaluation information matrix.	β	Sensitivity coefficient, $\beta \in [0, 1]$.
$\bar{z}_{ij}^k$	The Linguistic Z-numbers after the transformation of the linguistic scale function.	$\theta_i^k, \bar{\theta}_i$	The utility value of alternative a_i for DM d_k , the average utility value of alternative.
$\hat{Z}^k$	The evaluation information matrix after the transformation of the language scale function	γ	The traditional grey relational degree.
$\bar{z}_{ij}^k$	The crisp number derived from the conversion of linguistic Z-numbers.	ϵ	The resolution coefficient.
$\bar{Z}^k$	Crisp number evaluation information matrix.	ξ_{kl}	The improved grey relational degree of this paper.
$\bar{z}^+$	The crisp Z-numbers corresponding to the best linguistic Z-numbers $z^+ = (A_{2g}, B_{2g})$.	θ_{kl}	The interference angle between DM d_k and DM d_l .

these methods often ignore the psychological sensitivity and cognitive characteristics of DMs to changes in different attribute's weights.

Therefore, the paper establishes a novel multi-attribute quantum group decision-making (MAQGDM) model. Firstly, this paper assigns weights to DMs using regret theory, and the impact of this approach is systematically analysed. Secondly, a utility function integrating DMs' psychological preferences is formulated, and an optimization model is constructed to determine attribute's weights for each DM. On this basis, to address interference effects among DMs, a QLBN is constructed. In this network, the DMs' weights are taken as the first layer probability, while the conditional probability is calculated by attribute's weights and decision unit weight. Meanwhile, grey relational degree is innovatively introduced to measure the interference angle. Finally, the ranking of each alternative is completed by calculating its quantum probability. This paper presents the following contributions:

(1) Interference angle is measured based on grey relational degree. This method can effectively deal with decision-making scenarios with high data uncertainty and incomplete information. It does not require a typical probability distribution or a large number of samples, and can effectively handle nonlinear relations, capture complex correlations between variables, and fit the nonlinear characteristics of quantum phenomena.

(2) The weights of DMs are determined based on regret theory. Considering the preference differences among different DMs, the psychological feelings of DMs when adjusting feedback suggestions are quantified based on the regret-rejoice function, and the potential regrets of each DM are quantified, so as to reasonably assign the weight of each DM.

(3) An exponential preference utility function is proposed to describe the nonlinear psychological response of DMs to the difference between rating value and group rating value. On this basis, a goal optimization model is constructed to determine the optimal attribute's weights for DMs based on the maximum deviation principle.

The structure of this paper is organized as follows: The second section provides a review of linguistic scale function (LSF), linguistic Z-numbers, regret theory, grey relational degree and quantum probability theory. Section 3 proposes a MAQGDM model that incorporates psychological preferences of DMs. Section 4 validates the proposed model through a case study, with sensitivity and comparative analyses conducted to assess its robustness and stability. Finally, the fifth section provides a summary of the research contributions and proposes future research directions.

2. Preliminaries

For ease of understanding, the summary of the symbols used in this paper is shown in Table 1.

2.1. Linguistic scale function (LSF)

Definition 1. (van Hoef et al., 2023). Assuming there is a set of linguistic terms $L = \{l_0, l_1, \dots, l_{2g}\}$, where g is a positive integer, the LSF H mapping from l_c to $\zeta_c (c = 0, 1, \dots, 2g)$ is defined as follows.

$$H : l_c \rightarrow \zeta_c (c = 0, 1, \dots, 2g). \quad (1)$$

where symbol ζ_c reflects the DM's preference for linguistic term l_c , and $0 \leq \zeta_0 < \zeta_1 < \dots < \zeta_{2g} \leq 1$.

Definition 2. (Wang et al., 2014). The LSF $H_o(l_c) (o = 1, 2)$ is defined as follows:

$$H_1(l_c) = \left(\frac{c}{2g}\right)^{\frac{1}{\sigma}}, 0 \leq c \leq 2g. \quad (2)$$

$$H_2(l_c) = \begin{cases} \frac{g^\nu - (g-c)^\nu}{2g^\nu}, & 0 \leq c \leq g \\ \frac{g^\tau + (c-g)^\tau}{2g^\tau}, & g+1 \leq c \leq 2g. \end{cases} \quad (3)$$

where, $\nu, \tau \in (0, 1]$.

2.2. Z-numbers and linguistic Z-numbers

Z-number is a tool that can express both probability information and uncertainty degree. Zadeh pioneered the Z-number in 2011 (Zadeh, 2011), which establishes a novel mathematical paradigm for the structured representation of fuzzy information in uncertain environments.

Definition 3 (Ranjbar et al., 2022). Suppose there are two sets of linguistic terms $L_1 = \{l_0, l_1, \dots, l_{2g}\}$ and $L_2 = \{l'_0, l'_1, \dots, l'_{2g'}\}$, where g is a positive integer, and the linguistic Z-numbers comprises two components: a linguistic term $A (A \in L_1)$ that constrains the possible value range of an uncertain variable, and a linguistic term $B (B \in L_2)$ that quantifies its reliability degree.

Definition 4 (Wang, Ju, et al., 2024). Let $z_1 = (A_1, B_1)$ and $z_2 = (A_2, B_2)$ be two Z-numbers, then the weighted Euclidean distance between z_1 and z_2 is as follows.

$$d(z_1, z_2) = \sqrt{\alpha |f^*(A_1) - f^*(A_2)|^2 + \sigma |g^*(B_1) - g^*(B_2)|^2}. \quad (4)$$

where, α and σ represent the weights of the two components of the linguistic Z-numbers, respectively, to distinguish the importance of the two. And $0 \leq \sigma \leq \alpha \leq 1$, $\alpha + \sigma = 1$.

2.3. Quantum decision theory

Quantum probability theory is a mathematical theory that breaks away from the framework of classical probability theory and studies the probability laws of events occurring in quantum systems. It can be used to explain complex decision behaviour and uncertainty phenomena. Based on the Hilbert space framework of quantum mechanics, the theory describes the state and evolution of the system through state vectors and operators. Event probabilities are determined by the modulus squared of quantum state probability amplitudes, and the probability amplitude of different paths can interfere with each other, resulting in a non-classical probability distribution (Yan, Xu, et al., 2024). Assuming the existence of path selection T and path selection F , when the decision path is unknown, the probability distribution of conscious choice is as follows:

$$\begin{aligned} P &= |\psi_{TF}|^2 = |\psi_T + \psi_F|^2 = (\psi_T + \psi_F)(\psi_T + \psi_F)^* \\ &= \psi_T \psi_T^* + \psi_F \psi_F^* + \psi_T \psi_F^* + \psi_F \psi_T^* \\ &= |\psi_T|^2 + |\psi_F|^2 + \psi_T \psi_F^* + \psi_F \psi_T^* \\ &= |\psi_T|^2 + |\psi_F|^2 + 2\sqrt{|\psi_T|^2 \cdot |\psi_F|^2} \cos \theta_{TF} \end{aligned} \quad (5)$$

where, ψ_T and ψ_F represent the probability amplitudes of potential paths T and F respectively, ψ_T^* and ψ_F^* are the complex conjugations of them, while $\psi_T \psi_F^*$ and $\psi_F \psi_T^*$ are interference terms. θ_{TF} is the interference angle between these two paths.

2.4. Regret theory

The regret theory was founded by Bell (1982), Loomes and Sugden (1982), which pointed out that DMs would compare different results when faced with different choices. When the result of the selected alternative is worse than that of the unselected alternative, they would feel regret, and when they found that the selected alternative yielded superior outcomes to the unselected one, they would feel happy. This theory differs from the traditional decision-making theory which assumes that DMs are completely rational and only pursue the maximization of utility when making decisions. Instead, it fully takes into account the psychological differences of DMs when choosing one alternative and giving up other alternatives. This psychological difference may encourage DMs to consider more than expected benefits when making decisions, and to avoid making choices they will regret. The regret

perceived utility function consists of the utility function of the selected alternative and the regret-rejoice function:

$$U(a_1, a_2) = u(a_1) + R(u(a_1) - u(a_2)). \quad (6)$$

where $u(a_1)$ and $u(a_2)$ are the utility values generated by alternative a_1 and a_2 respectively, $R(u(a_1) - u(a_2))$ is the regret-rejoice value and $R = 1 - e^{-\delta(u(a_1) - u(a_2))}$, when $R(u(a_1) - u(a_2)) \geq 0$, it indicates that the utility of alternative a_1 is greater than that of alternative a_2 , and the DM is happy. When $R(u(a_1) - u(a_2)) \leq 0$, it indicates that the utility of alternative a_1 is less than that of alternative a_2 , and the DM feels regret.

The regret perceived utility function is defined as:

$$U(a_i) = u(a_i) + R(u(a_i) - u^*(a_i)), i = 1, 2, \dots, m. \quad (7)$$

where, $U(a_i)$ represents the utility generated by alternative a_i ($i = 1, 2, \dots, m$), $R(\cdot)$ is the regret-rejoice function, which is a monotonically increasing concave function, $R(u(a_i) - u^*(a_i)) \leq 0$ represents the regret value of choosing alternative a_i , while $u^*(a_i)$ represents the highest utility that can be obtained among all alternatives.

2.5. Grey relational degree

Grey relational degree, as the core metric in grey relational analysis, is theoretically grounded in quantifying relational intensity through geometric similarity comparison between sequence curves. The relational degree between the corresponding sequences exhibits a monotonically increasing relationship with curve proximity.

Definition 5. (Liu & Forrest, 2010). Suppose there is a reference sequence: $X_0 = \{x_0(1), x_0(2), \dots, x_0(r)\}$ and a comparison sequence: $X_k = \{x_k(1), x_k(2), \dots, x_k(r)\}$, where $r \geq 2$ and $k = 1, 2, \dots, s$.

The grey relational coefficient between the comparison sequence and the reference sequence is:

$$\gamma(x_0(i), x_k(i)) = \frac{\min_i \min_k |x_0(i) - x_k(i)| + \epsilon \max_i \max_k |x_0(i) - x_k(i)|}{|x_0(i) - x_k(i)| + \epsilon \max_i \max_k |x_0(i) - x_k(i)|}. \quad (8)$$

where, ϵ is the resolution coefficient and $\epsilon \in (0, 1)$. On this basis, the grey relational degree between the comparison sequence and the reference sequence is as follows:

$$\gamma(X_0, X_k) = \frac{1}{r} \sum_{i=1}^r \gamma(x_0(i), x_k(i)). \quad (9)$$

Grey relational degree $\gamma(X_0, X_k)$ takes values in the interval $(0, 1]$.

3. A MAGDM model considering quantum interference effects

In the context of a MAGDM problem, let there be alternatives denoted as $A' = \{a_1, a_2, \dots, a_m\}$ ($m \geq 2$), n attributes represented by $C = \{c_1, c_2, \dots, c_n\}$ ($n \geq 2$), and a DM group consisting of s individuals expressed as $D = \{d_1, d_2, \dots, d_s\}$ ($s \geq 2$). The attribute's weights under different DMs' evaluations are unknown, and need to be determined based on the evaluation matrices. The set of attribute's weights for each DM is recorded as $W^k = \{\omega_1^k, \omega_2^k, \dots, \omega_n^k\}$, where $0 \leq \omega_j^k \leq 1$, and the sum of weights for each DM satisfies $\sum_{j=1}^n \omega_j^k = 1$, ($k = 1, 2, \dots, s$). The set of DMs' weights is denoted as $\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_s\}$, where $0 \leq \lambda_k \leq 1$, and the sum of all DMs' weights satisfies $\sum_{k=1}^s \lambda_k = 1$.

3.1. Collection and processing of decision information

(1) The evaluation information of alternative a_i ($i = 1, 2, \dots, m$) about attribute c_j ($j = 1, 2, \dots, n$) represented by linguistic Z-numbers of each DM is collected. The evaluation information matrix Z^k ($k = 1, 2, \dots, s$) is as follows:

$$Z^k = \begin{pmatrix} z_{11}^k & z_{12}^k & \dots & z_{1n}^k \\ z_{21}^k & z_{22}^k & \dots & z_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1}^k & z_{m2}^k & \dots & z_{mn}^k \end{pmatrix}, k = 1, 2, \dots, s. \quad (10)$$

Among them, the initial evaluation information is represented by the linguistic Z-numbers $z_{ij}^k = (A_{ij}^k, B_{ij}^k)$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$). On this basis, the evaluation information matrix $Z^k = (z_{ij}^k)_{m \times n}$ of each DM is transformed into linguistic scale function matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m \times n} = [f^*(A_{ij}^k), g^*(B_{ij}^k)]_{m \times n}$ by Eqs. (2)–(3), where $f^*(A_{ij}^k) = H_1(A_{ij}^k)$ and $g^*(B_{ij}^k) = H_2(B_{ij}^k)$.

(2) After obtaining the language scale function matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m \times n}$, it is necessary to convert it into a crisp number evaluation information matrix to facilitate subsequent calculations. Scholars usually convert linguistic Z-numbers into fuzzy numbers (Xiao et al., 2026) or precise numbers (Chai et al., 2023; Chen et al., 2023; Du et al., 2024; Wang, Ju, et al., 2024) for processing. This paper adopts the linguistic Z-number conversion method proposed by Wang, Ju, et al. (2024), which is the latter type of clear processing method. This further optimizes the information retention effect: it converts the evaluation part A and the reliability part B of the linguistic Z-number into crisp numerical values through the linguistic scale function that meets the order requirement. Following the idea of relative proximity, these values are then transformed into a single crisp evaluation value based on the distance between the linguistic Z-number and the positive and negative ideal Z-numbers. This conversion method does not need to convert the linguistic Z-number into intermediate forms such as triangular fuzzy numbers, and directly performs numerical calculations, effectively avoiding information loss in the intermediate conversion process (Wang, Ju, et al., 2024). It completely retains the core information of the evaluation result and the reliability of the information, so this conversion method is a reasonable choice that balances information integrity and computational feasibility, and can meet the decision-making requirements of this paper. To retain more information and enhance decision-making accuracy, this paper measures the closeness degree of decision units by computing their relative distances to both the positive and negative ideal solutions (Wang, Ju, et al., 2024).

On this basis, linguistic Z-numbers can be converted into crisp numbers through the following calculation (Wang, Ju, et al., 2024):

$$\hat{z}_{ij}^k = \frac{d(\tilde{z}_{ij}^k, \tilde{z}^+)^2}{d(\tilde{z}_{ij}^k, \tilde{z}^+)^2 + d(\tilde{z}_{ij}^k, \tilde{z}^-)^2}, i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (11)$$

The crisp number evaluation information matrix for each DM is denoted as $\hat{Z}^k = (\hat{z}_{ij}^k)_{m \times n}$, where $\tilde{z}^+$ represents the best Z-numbers corresponding to the form of the best linguistic Z-numbers $z^+ = (A_{2g}, B_{2g'})$, while $\tilde{z}^-$ denotes the worst Z-numbers corresponding to the form of the worst linguistic Z-numbers $z^- = (A_0, B_0)$.

3.2. DMs' weights determination based on regret theory

In the process of MAGDM, the preferences of different DMs are often different, and the realization of group consistency is faced with great challenges. This paper adopts the idea of regret theory, quantifies DMs' psychological feelings when adjusting feedback suggestions by constructing regret–rejoice function, quantifies each DM's potential regret, and measures their determination of the evaluation of the program. The stronger the regret, the more wavering the DM is when making the evaluation, so the weight in the decision should be correspondingly smaller. In this way, the weight of each DM in the decision is reasonably allocated.

Definition 6 (Loomes & Sugden, 1982). $\hat{z}_{ij}^k$ is an element in crisp number evaluation information matrix $\hat{Z}^k = (\hat{z}_{ij}^k)_{m \times n}$ of DM d_k . The regret–rejoice function μ_{ij}^k is defined as:

$$\mu_{ij}^k = 1 - EXP \left(-\delta \left(\left(\max_k \hat{z}_{ij}^k \right)^p - \left(\hat{z}_{ij}^k \right)^p \right) \right), \quad (12)$$

$i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s.$

where, δ ($\delta = 0.3$) is regret aversion coefficient. p ($p = 0.88$) is risk aversion coefficient (Shen et al., 2020).

On this basis, the regret matrix of each DM is calculated as follows:

$$R^k = \begin{pmatrix} \mu_{11}^k & \mu_{12}^k & \cdots & \mu_{1n}^k \\ \mu_{21}^k & \mu_{22}^k & \cdots & \mu_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{m1}^k & \mu_{m2}^k & \cdots & \mu_{mn}^k \end{pmatrix}, k = 1, 2, \dots, s. \quad (13)$$

Secondly, calculate the regret utility value of each DM for each alternative:

$$q_i^k = \sum_{j=1}^n \omega_j^0 \mu_{ij}^k, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (14)$$

where ω_j^0 represents the initial attribute's weights assigned subjectively.

On this basis, the group minimum regret utility value of each alternative is defined as:

$$q_i^* = \min_{1 \leq k \leq s} q_i^k, i = 1, 2, \dots, m. \quad (15)$$

where, the difference between each DM and the group's minimum regret utility is calculated as follows:

$$\Delta q_i^k = q_i^k - q_i^*, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (16)$$

Since Δq_i^k may be 0, a deviation term 1 is introduced, and the weight calculation formula for each DM is constructed on this basis, as follows:

$$\lambda_k = \frac{1 / \sum_{i=1}^m (1 + \Delta q_i^k)}{\sum_{k=1}^s (1 / \sum_{i=1}^m (1 + \Delta q_i^k))}, k = 1, 2, \dots, s. \quad (17)$$

Theorem 1. λ_k decreases with the increase of Δq_i^k .

Proof. Firstly, let $S_k = \sum_{i=1}^m (1 + \Delta q_i^k)$ ($k = 1, 2, \dots, s$), $S_k > 0$ and S_k increases with Δq_i^k , so $\lambda_k = \frac{1/S_k}{\sum_{k=1}^s (1/S_k)}$. Secondly, the first derivative of λ_k with respect to S_k is $-\frac{1/S_k}{S_k^2 \cdot (\sum_{k=1}^s (1/S_k))^2}$, and since $1/S_k < \sum_{k=1}^s (1/S_k)$ ($k > 1$) and $(S_k)^2 \cdot (\sum_{k=1}^s (1/S_k))^2 > 0$, $-\frac{1/S_k}{S_k^2 \cdot (\sum_{k=1}^s (1/S_k))^2} < 0$, it follows that λ_k decreases with the increase of S_k . And because S_k increases as Δq_i^k increases, λ_k decreases with the increase of Δq_i^k . $\square$

Corollary 1. According to Theorem 1, when the regret value of each DM for a certain alternative is closer to the minimum regret value of the group for the same alternative, that is, the smaller the value of Δq_i^k , the smaller the regret degree of each DM, and thus the greater decision weight is assigned to him.

3.3. Determination of attributes' weights considering preference utility

In MAGDM problems, the weights assigned by DMs to each attribute are usually influenced by personal preferences, experience, and changes in the external environment, resulting in a certain degree of dynamicity in weight satisfaction. However, traditional utility functions still have significant limitations in depicting the aforementioned psychological behavioural characteristics: Firstly, the psychological responses of DMs often exhibit nonlinear features, and traditional linear functions are often unable to capture this (Mi et al., 2019). Secondly, some nonlinear functions, such as $U(r) = \frac{1-e^{-\alpha r}}{\alpha}$, where r represents the evaluation value of an alternative and α ($\alpha \in (0, 1)$) is the risk aversion parameter, often assume that marginal utility is decreasing (Wen et al., 2024). This characteristic of marginal utility decreasing may cause DMs to underestimate the value of high-potential solutions, which is not conducive to the identification and selection of innovative or

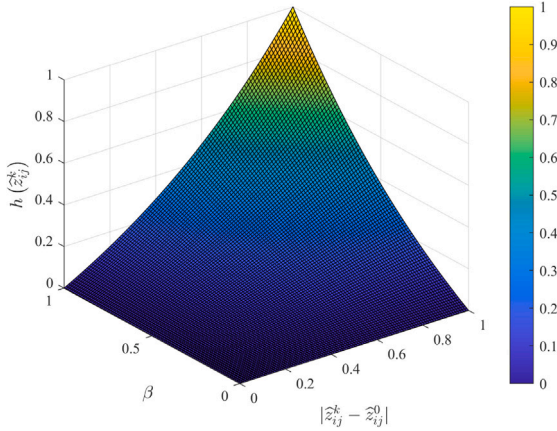


Fig. 1. A novel preference utility function.

breakthrough alternatives. To address these issues, this paper constructs a new type of psychological preference function, which has the following characteristics: Firstly, a tunable parameter β is designed, which can continuously depict the sensitivity of decision-makers to evaluation differences, enhance the model's adaptability to different psychological characteristics, and also be able to depict the nonlinear characteristics of DMs' psychological responses. Second, its marginal utility is gradually increasing, and this characteristic of marginal utility increasing helps to amplify the utility differences, thereby prompting DMs to adjust the scores that deviate significantly or adhere to valuable and differentiated viewpoints.

Definition 7. Suppose $\hat{z}_{ij}^0$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n$) represents the reference value of the group's evaluation of alternative a_i on attribute c_j . The psychological preference utility is defined as:

$$h\left(\hat{z}_{ij}^k\right) = \frac{1 - e^{-\beta \left| \hat{z}_{ij}^k - \hat{z}_{ij}^0 \right|}}{1 - e}, i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (18)$$

where, $\hat{z}_{ij}^0 = \sum_{k=1}^s \lambda_k \hat{z}_{ij}^k$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$), β ($\beta \in [0, 1]$) is the sensitivity coefficient, which controls the reaction rate of DMs to changes in the difference between the evaluation value and the group evaluation benchmark value. If β is large, the utility value increases rapidly with the increase of the difference. If β is small, the utility value increases more gently. By adjusting β , it is possible to control the sensitivity of the DM to changes in the difference, thus simulating its preferred behaviour more accurately. The utility function is graphically illustrated in Fig. 1.

Theorem 2. The utility function $h\left(\hat{z}_{ij}^k\right)$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) is a convex function.

Proof. Firstly, taking the first derivative of the utility function yields: when $\hat{z}_{ij}^k \geq \hat{z}_{ij}^0$, $\frac{dh\left(\hat{z}_{ij}^k\right)}{d\hat{z}_{ij}^k} = \frac{-\beta e^{\beta\left(\hat{z}_{ij}^k - \hat{z}_{ij}^0\right)}}{1 - e}$, where $\hat{z}_{ij}^0 = \sum_{k=1}^s \lambda_k \hat{z}_{ij}^k$ is a real number. Secondly, taking the second derivative of the utility function $h\left(\hat{z}_{ij}^k\right)$ yields: $\frac{d^2h\left(\hat{z}_{ij}^k\right)}{d^2\hat{z}_{ij}^k} = \frac{-\beta^2 e^{\beta\left(\hat{z}_{ij}^k - \hat{z}_{ij}^0\right)}}{1 - e}$, since $1 - e < 0$ and $-\beta^2 e^{\beta\left(\hat{z}_{ij}^k - \hat{z}_{ij}^0\right)} \leq 0$, $\frac{-\beta^2 e^{\beta\left(\hat{z}_{ij}^k - \hat{z}_{ij}^0\right)}}{1 - e} \geq 0$, that is, when $\hat{z}_{ij}^k \geq \hat{z}_{ij}^0$, the utility function $h\left(\hat{z}_{ij}^k\right)$ is a convex function. When $\hat{z}_{ij}^k < \hat{z}_{ij}^0$, the second derivative of the utility function $h\left(\hat{z}_{ij}^k\right)$ yields: $\frac{d^2h\left(\hat{z}_{ij}^k\right)}{d^2\hat{z}_{ij}^k} =$

$\frac{-\beta^2 e^{\beta\left(\hat{z}_{ij}^k - \hat{z}_{ij}^0\right)}}{1 - e}$, and the utility function $h\left(\hat{z}_{ij}^k\right)$ is a convex function. In summary, the utility function is a convex function. $\square$

Corollary 2. With the increase of $\hat{z}_{ij}^k - \hat{z}_{ij}^0$, the increase speed of the DM's preference utility value gradually accelerates. In other words, when the DM's rating deviates from the group value benchmark, the utility change caused by the unit deviation will accelerate. This feature of increasing marginal utility helps to achieve the consistency of group decision making, and through the amplification effect of utility differences, DMs can actively adjust the obviously deviated scores or insist on valuable differentiation views.

Based on this, considering the idea of maximizing deviation (Farokhzadeh et al., 2021; Rani & Mishra, 2021), this paper constructs a goal optimization model, aiming to solve the weights of each attribute for each DM according to the preference utility function of the DMs. Specifically, the optimization goal of this paper is to optimize the allocation of attribute's weights by maximizing the sum of the squared deviations between the utility of each alternative and the group preference value. The model is as follows:

$$\begin{aligned} M_1 : \max & \sum_{k=1}^s \sum_{i=1}^m \left(\sum_{j=1}^n \omega_j^k h\left(\hat{z}_{ij}^k\right) - \sum_{k=1}^s \lambda_k \sum_{j=1}^n \omega_j^k h\left(\hat{z}_{ij}^k\right) \right)^2 \\ \text{s.t.} & \begin{cases} h\left(\hat{z}_{ij}^k\right) = \frac{1 - e^{-\beta \left| \hat{z}_{ij}^k - \hat{z}_{ij}^0 \right|}}{1 - e} & (19-1) \\ \sum_{j=1}^n \omega_j^k = 1 & (19-2) \\ 0 \leq \omega_j^k \leq 1, \beta \in [0, 1] & (19-3) \\ \hat{z}_{ij}^0 = \sum_{k=1}^s \lambda_k \hat{z}_{ij}^k & (19-4) \\ i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s & (19-5) \end{cases} \end{aligned} \quad (19)$$

Theorem 3. The Model (19) has an optimal solution.

Proof. Firstly, the objective function is a continuous function of ω_j^k and $\hat{z}_{ij}^0$, and the constraint (19-1) is continuous; secondly, the variable ω_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$) takes on the interval $[0, 1]$, and $\sum_{j=1}^n \omega_j^k = 1$, so that the feasible domain is closed and bounded. Therefore, according to Weierstrass limit theorem (Zhang et al., 2024), a continuous function must have a maximum and a minimum on a closed and bounded feasible field. Therefore, the existence of an optimal solution to Model (19) is provable. $\square$

3.4. Ranking alternatives based on quantum probability

3.4.1. Construction of a QLBN

Firstly, let the DMs' weights be the first layer probability of QLBN:

$$P(d_k) = \lambda_k, k = 1, 2, \dots, s. \quad (20)$$

Secondly, conditional probability in QLBN is calculated as follows:

$$P(a_i | d_k) = \frac{\sum_{j=1}^n \omega_j^k \hat{z}_{ij}^k}{\sum_{j=1}^n \omega_j^k \hat{z}_{ij}^k}, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (21)$$

3.4.2. Interference angle measure based on grey relational degree

Firstly, build vectors $\bar{X}_k$ ($k = 1, 2, \dots, s$) and $\bar{Y}_l$ ($l = 1, 2, \dots, s$):

$$\bar{X}_k = \begin{bmatrix} x_1^k \\ x_2^k \\ \vdots \\ x_m^k \end{bmatrix}, \bar{Y}_l = \begin{bmatrix} y_1^l \\ y_2^l \\ \vdots \\ y_m^l \end{bmatrix}, \quad (22)$$

$k = 1, 2, \dots, s, l = 1, 2, \dots, s, l \neq k.$

where,

$$\begin{aligned} x_i^k &= P(d_k)P(a_i|d_k), i = 1, 2, \dots, m, k = 1, 2, \dots, s, \\ y_i^l &= P(d_l)P(a_i|d_l), i = 1, 2, \dots, m, l = 1, 2, \dots, s, l \neq k. \end{aligned} \quad (23)$$

On this basis, the difference is defined as:

$$\begin{aligned} \Delta x_i^k &= x_{i+1}^k - x_i^k, i = 1, 2, \dots, m-1, k = 1, 2, \dots, s, \\ \Delta y_i^l &= y_{i+1}^l - y_i^l, i = 1, 2, \dots, m-1, l = 1, 2, \dots, s, l \neq k. \end{aligned} \quad (24)$$

Secondly, construct the difference sequence vector:

$$\begin{aligned} \overline{\Delta x_i^k} &= \begin{bmatrix} P(d_k)P(a_2|d_k) - P(d_k)P(a_1|d_k) \\ P(d_k)P(a_3|d_k) - P(d_k)P(a_2|d_k) \\ \vdots \\ P(d_k)P(a_m|d_k) - P(d_k)P(a_{m-1}|d_k) \end{bmatrix}, \\ \overline{\Delta y_i^l} &= \begin{bmatrix} P(d_l)P(a_2|d_l) - P(d_l)P(a_1|d_l) \\ P(d_l)P(a_3|d_l) - P(d_l)P(a_2|d_l) \\ \vdots \\ P(d_l)P(a_m|d_l) - P(d_l)P(a_{m-1}|d_l) \end{bmatrix}, \end{aligned} \quad (25)$$

$k = 1, 2, \dots, s, l = 1, 2, \dots, s, l \neq k.$

In the calculation of the grey relational degree between DM d_k and DM d_l , $\overline{\Delta x_i^k}$ is the reference sequence and $\overline{\Delta y_i^l}$ is the comparison sequence.

Finally, the grey relational degree between the two vectors is defined as:

$$\begin{aligned} \xi_{kl} &= \frac{1}{m-1} \sum_{i=1}^{m-1} \text{sgn}(\Delta x_i^k, \Delta y_i^l) \cdot \gamma(\Delta x_i^k, \Delta y_i^l) \\ &= \frac{1}{m-1} \sum_{i=1}^{m-1} \text{sgn}(\Delta x_i^k, \Delta y_i^l) \cdot \frac{\min_i |\Delta x_i^k - \Delta y_i^l| + \epsilon \max_i |\Delta x_i^k - \Delta y_i^l|}{|\Delta x_i^k - \Delta y_i^l| + \epsilon \max_i |\Delta x_i^k - \Delta y_i^l|}, \end{aligned} \quad (26)$$

$$k = 1, 2, \dots, s, l = 1, 2, \dots, s, k \neq l.$$

The grey relational degree can describe the degree of correlation between DMs. This relational degree can reflect the extent of mutual influence among DMs. In reality, not only positive correlations exist between DMs, but also negative correlations. The traditional grey relational degree $\gamma \in (0, 1]$ can only measure the similarity between sequences and cannot reflect the direction information of the changing trend, making it difficult to reflect the direction of mutual influence among DMs. Therefore, in this paper, based on Eq. (8), the sgn function is introduced to construct Eq. (26) to extend the grey relational to the range of $[-1, 0) \cup (0, 1]$ (Liu, 2023), so that the degree of influence between DMs can be measured while the direction of mutual influence can also be reflected.

$$\text{sgn}(\Delta x_h, \Delta y_h) = \begin{cases} 1 & , \text{if } \Delta x_h \cdot \Delta y_h \geq 0 \\ -1 & , \text{if } \Delta x_h \cdot \Delta y_h < 0. \end{cases} \quad (27)$$

Theorem 4. The value range of grey relational degree is $[-1, 0) \cup (0, 1]$.

Proof. Firstly, the value range of the traditional grey relational coefficient $\gamma(\Delta x_h, \Delta y_h)$ is $(0, 1]$; secondly, the value of $\text{sgn}(\Delta x_h, \Delta y_h)$ is $\{-1, 1\}$; therefore, when $\Delta x_h \cdot \Delta y_h < 0$, the grey relational coefficient of the two vectors is positive; when $\Delta x_h \cdot \Delta y_h > 0$, the grey relational coefficient of the two vectors is negative; therefore, the value range of grey relational degree ξ_{kl} is $[-1, 0) \cup (0, 1]$. $\square$

The combination of the grey relational degree and the quantum decision framework is based on the fact that both can depict the correlation and mutual influence between DMs in terms of trend judgement. The grey relational degree is used to measure the degree of closeness of the DM evaluation sequence in terms of the changing trend and geometric shape, and can stably depict the potential correlation strength between DMs under conditions of small samples and incomplete information. Its value range is $[-1, 0) \cup (0, 1]$, and this correlation strength can reflect the direction and intensity of the mutual influence between DMs.

Definition 8. Let θ_{kl} be the interference angle between DM d_k ($k = 1, 2, \dots, s$) and DM d_l ($l = 1, 2, \dots, s, k \neq l$). The cosine value $\cos(\theta_{kl})$ of the interference angle is defined as:

$$\cos(\theta_{kl}) = \xi_{kl}, k = 1, 2, \dots, s, l = 1, 2, \dots, s, k \neq l. \quad (28)$$

The quantum decision theory characterizes the interference effect between DM d_k ($k = 1, 2, \dots, s$) and DM d_l ($l = 1, 2, \dots, s, k \neq l$) through the measurement of the interference angle θ_{kl} , and the value range of the interference angle θ_{kl} is $[0, \pi]$. When calculating quantum probability, the value range of $\cos(\theta_{kl})$ is $[-1, 1]$, and when $\theta_{kl} = \frac{\pi}{2}$, $\cos(\theta_{kl}) = 0$. At this time, DMs are independent of each other and there is no mutual interference. The expanded grey relational degree in this paper has its own value range of $[-1, 0) \cup (0, 1]$. Moreover, the magnitude of the absolute value of the grey relational degree ξ_{kl} can also reflect the intensity of the interference effect between the two DMs. The absolute value reflects the consistency or opposition intensity of the two DMs in judging the trend. The more similar or opposite the trends are, the larger the absolute value of the relational degree is, indicating that the mutual influence between the two in the decision-making process is more significant. Therefore, defining the cosine value of the interference angle $\cos(\theta_{kl})$ as the grey relational degree ξ_{kl} not only provides a computable representation method for the abstract concept of “interference effect”, but also enhances the applicability and interpretability of the model when dealing with uncertain decision-making problems.

Theorem 5. When the grey relational degree $\xi_{kl} \in (0, 1]$, there is positive interference between DM d_k ($k = 1, 2, \dots, s$) and DM d_l ($l = 1, 2, \dots, s, k \neq l$). When the grey relational degree $\xi_{kl} \in [-1, 0)$, there is negative interference between DM d_k and DM d_l .

Proof. Firstly, as can be seen from Eq. (29), when $\Delta x_h \cdot \Delta y_h \geq 0$, either $\Delta x_h \geq 0$ and $\Delta y_h \geq 0$, or $\Delta x_h \leq 0$ and $\Delta y_h \leq 0$, the trend directions of the two vectors are consistent. The direction of interference between DM d_k and DM d_l is positive, that is, $\cos(\theta_{kl}) > 0$. At this time, since the value range of the traditional grey relational coefficient $\gamma(\Delta x_h, \Delta y_h)$ is $(0, 1]$, $\text{sgn}(\Delta x_h, \Delta y_h) = 1$, therefore, the grey relational degree $\xi_{kl} \in (0, 1]$. That is, when the grey relational degree $\xi_{kl} \in (0, 1]$, the trend directions of the two vectors are consistent, and there is positive interference between DM d_k and DM d_l . Similarly, when the grey relational degree $\xi_{kl} \in [-1, 0)$, the trend directions of the two vectors are opposite, and there is negative interference between DM d_k and DM d_l . $\square$

Remark 1. For a problem involving s DMs, the method based on Deng entropy measure for calculating the interference angle requires calculating the ordered pairing correlations among all DMs, and its complexity is $O(s^2)$. This complexity will increase significantly as the number of DMs increases. The method proposed in this paper to measure the interference angle by using the grey relational degree has the core influencing factor of the number of DMs for its calculation complexity. The process of calculating the grey relational degree between different DMs needs to cover all pairing combinations of the decision-makers. As the number of DMs s increases, the number of interference effects will increase significantly. That is, when the number of DMs is s , the complexity of this method is $O\left(\frac{s^2}{2}\right)$, which is much smaller than the measurement method based on Deng entropy.

3.4.3. Calculate the quantum probabilities of each alternative

Calculate the quantum probability of each alternative:

$$P(a_i) = \frac{1}{\rho} \left(\begin{aligned} & \sum_{k=1}^s P(d_k) P(a_i|d_k) \\ & + 2 \sum_{k=1}^{s-1} \sum_{l=k+1}^s \sqrt{P(d_k) P(a_i|d_k) P(d_l) P(a_i|d_l)} \cos(\theta_{kl}) \end{aligned} \right), \quad (29)$$

$i = 1, 2, \dots, m.$

where,

$$\rho = \sum_{i=1}^m \left(\frac{\sum_{k=1}^s P(d_k) P(a_i|d_k)}{+2 \sum_{k=1}^{s-1} \sum_{l=k+1}^s \sqrt{P(d_k) P(a_i|d_k) P(d_l) P(a_i|d_l)} \cos(\theta_{kl})} \right) \quad (30)$$

3.5. Decision steps

In this paper, we propose a MAQGDM model considering DMs' preferences. The decision steps are as follows, as shown in Fig. 2.

Step 1: Collect evaluation information. Combine the evaluation information of DM d_k ($k = 1, 2, \dots, s$) on alternative a_i ($i = 1, 2, \dots, m$) regarding attribute c_j ($j = 1, 2, \dots, n$) into k evaluation information matrices $Z^k = (z_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, s$).

Step 2: Process the evaluation information. Firstly, convert the evaluation information matrix $Z^k = (z_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, s$) into the linguistic scale function matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{m \times n} = [f^*(A_{ij}^k), g^*(B_{ij}^k)]_{m \times n}$ ($k = 1, 2, \dots, s$) through Eqs. (2)–(3). Finally, transform the evaluation information matrix into the crisp linguistic Z-numbers evaluation information matrix $\hat{z}_{ij}^k = (\hat{z}_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, s$) according to Eq. (11).

Step 3: Determine the DMs' weights based on regret theory. Firstly, obtain the regret matrix $R^k = (r_{ij}^k)_{m \times n}$ ($k = 1, 2, \dots, s$) of each DM through Eqs. (12)–(13). Secondly, calculate the regret utility value q_i^k ($i = 1, 2, \dots, m, k = 1, 2, \dots, s$) of each DM for each alternative and the difference Δq_i^k ($i = 1, 2, \dots, m, k = 1, 2, \dots, s$) between each DM and the group's minimum regret utility according to Eqs. (14)–(16). On this basis, calculate the weights λ^k ($k = 1, 2, \dots, s$) of each DM according to Eq. (17).

Step 4: Calculate the attribute's weights of each DM through the goal optimization model considering DMs' preferences. The attribute's weights ω_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$) of each DM are obtained by Model (19).

Step 5: Determine the quantum-like Bayesian network. Firstly, set the DMs' weights λ^k ($k = 1, 2, \dots, s$) as the first layer probability $P(d_k)$ ($k = 1, 2, \dots, s$) of the QLBN. Secondly, calculate the conditional probability $P(a_i|d_k)$ ($i = 1, 2, \dots, m, k = 1, 2, \dots, s$) of the QLBN according to Eq. (21).

Step 6: Calculate the cosine value of the interference angle based on grey relational degree. Firstly, construct two vectors X and Y composed of $P(d_k) P(a_i|d_k)$ ($i = 1, 2, \dots, m, k = 1, 2, \dots, s$) for DMs d_k ($k = 1, 2, \dots, s$) and d_l ($l = 1, 2, \dots, s, k \neq l$) according to Eqs. (22)–(23). Secondly, calculate the grey relational degree ξ_{kl} ($k = 1, 2, \dots, s, l = 1, 2, \dots, s, k \neq l$) between the two vectors according to Eqs. (24)–(27). Finally, calculate the cosine value $\cos(\theta_{kl})$ ($k = 1, 2, \dots, s, l = 1, 2, \dots, s, k \neq l$) of the interference angle according to Eq. (28).

Step 7: Evaluate each alternative based on quantum probability. Calculate the quantum probability $P(a_i)$ ($i = 1, 2, \dots, m$) of each alternative according to Eq. (29) and rank the alternatives based on the quantum probability.

4. Case analysis

4.1. Background

With the global manufacturing industry's transition towards intelligence and flexibility, a large-scale automotive manufacturing enterprise plans to introduce a new generation of intelligent logistics systems to optimize the real-time distribution process of components from the warehouse to the assembly line. The enterprise is confronted with the challenge of a multi-variety and small-batch production model, and the traditional logistics methods have become inadequate to meet the high-frequency and high-precision distribution demands, leading to

frequent issues such as production line material shortages, inventory overstock, and slow response. To enhance overall production efficiency and supply chain resilience, the enterprise has decided to select the best system from five candidate intelligent logistics equipment solutions $\{a_1, a_2, a_3, a_4, a_5\}$ for deployment. These solutions include: a_1 is a large multi-purpose autonomous guided vehicle, a_2 is a high-speed material handling robot, a_3 is a liftable shuttle vehicle, a_4 is a lightweight AGV, and a_5 is a low-altitude logistics drone. To ensure the scientificity and efficiency of the evaluation process, three DMs $\{d_1, d_2, d_3\}$ from different fields assess each alternative against five attributes $\{c_1, c_2, c_3, c_4, c_5\}$: arrival time, material transportation volume, system stability, coverage range, and operational flexibility.

4.2. Decision steps

Step 1: Collect evaluation information. The two linguistic sets for this case are as follows:

$$A = \{A_0 : \text{very poor}, A_1 : \text{poor}, A_2 : \text{fair}, A_3 : \text{good}, A_4 : \text{very good}\}$$

$$B = \left\{ \begin{array}{l} B_0 : \text{completely unlikely}, B_1 : \text{extremely unlikely}, B_2 : \text{very unlikely}, \\ B_3 : \text{unlikely}, B_4 : \text{moderately likely}, B_5 : \text{relatively likely}, \\ B_6 : \text{very likely}, B_7 : \text{extremely likely}, B_8 : \text{completely likely} \end{array} \right\}$$

On this basis, the evaluation information matrices of each DM are as follows:

$$Z^1 = \begin{bmatrix} (A_3, B_4) & (A_3, B_7) & (A_2, B_5) & (A_3, B_4) & (A_2, B_2) \\ (A_4, B_7) & (A_1, B_5) & (A_4, B_7) & (A_4, B_8) & (A_4, B_7) \\ (A_2, B_6) & (A_3, B_6) & (A_2, B_3) & (A_2, B_4) & (A_2, B_3) \\ (A_2, B_5) & (A_1, B_4) & (A_1, B_2) & (A_2, B_6) & (A_3, B_7) \\ (A_2, B_3) & (A_3, B_3) & (A_2, B_5) & (A_2, B_5) & (A_1, B_3) \end{bmatrix},$$

$$Z^2 = \begin{bmatrix} (A_3, B_4) & (A_4, B_8) & (A_2, B_4) & (A_3, B_3) & (A_3, B_2) \\ (A_4, B_6) & (A_1, B_4) & (A_4, B_6) & (A_4, B_7) & (A_4, B_8) \\ (A_2, B_4) & (A_3, B_5) & (A_1, B_6) & (A_2, B_5) & (A_3, B_2) \\ (A_3, B_3) & (A_1, B_6) & (A_2, B_5) & (A_3, B_6) & (A_4, B_6) \\ (A_2, B_4) & (A_3, B_6) & (A_3, B_2) & (A_3, B_3) & (A_2, B_5) \end{bmatrix},$$

$$Z^3 = \begin{bmatrix} (A_3, B_5) & (A_3, B_7) & (A_3, B_6) & (A_4, B_4) & (A_1, B_7) \\ (A_3, B_6) & (A_2, B_4) & (A_4, B_7) & (A_3, B_7) & (A_4, B_8) \\ (A_1, B_7) & (A_2, B_6) & (A_1, B_3) & (A_1, B_6) & (A_2, B_5) \\ (A_2, B_6) & (A_0, B_7) & (A_1, B_6) & (A_1, B_4) & (A_3, B_8) \\ (A_2, B_5) & (A_3, B_5) & (A_3, B_4) & (A_2, B_5) & (A_1, B_6) \end{bmatrix}.$$

Step 2: Firstly, the evaluation information matrix $Z^k = (z_{ij}^k)_{5 \times 5}$ ($k = 1, 2, 3$) is transformed into the LSF matrix $\tilde{Z}^k = (\tilde{z}_{ij}^k)_{5 \times 5} = [H_1(A_{ij}^k), H_2(B_{ij}^k)]_{5 \times 5}$ ($k = 1, 2, 3$) by Eqs. (2)–(3). In order to lose the generality, this paper takes $\alpha = 0.5$ and $\sigma = 0.5$ in Eq. (4). Then, the crisp linguistic Z-numbers evaluation information matrix is obtained by Eq. (11):

$$\hat{Z}^1 = \begin{bmatrix} 0.7887 & 0.9863 & 0.8775 & 0.7887 & 0.3904 \\ 0.9976 & 0.7222 & 0.9976 & 1.0000 & 0.9976 \\ 0.9197 & 0.9740 & 0.4646 & 0.6907 & 0.4646 \\ 0.8775 & 0.5000 & 0.2172 & 0.9197 & 0.9863 \\ 0.4646 & 0.5833 & 0.3153 & 0.8775 & 0.2778 \end{bmatrix},$$

$$\hat{Z}^2 = \begin{bmatrix} 0.7887 & 1.0000 & 0.6907 & 0.5833 & 0.5082 \\ 0.9877 & 0.5000 & 0.9877 & 0.9976 & 1.0000 \\ 0.6907 & 0.9422 & 0.7828 & 0.8775 & 0.5082 \\ 0.5833 & 0.7828 & 0.8775 & 0.9740 & 0.9877 \\ 0.6907 & 0.9740 & 0.5082 & 0.5833 & 0.8775 \end{bmatrix},$$

$$\hat{Z}^3 = \begin{bmatrix} 0.9422 & 0.9863 & 0.9740 & 0.8333 & 0.8149 \\ 0.9740 & 0.6907 & 0.9976 & 0.9863 & 1.0000 \\ 0.8149 & 0.9197 & 0.2778 & 0.7828 & 0.8775 \\ 0.9197 & 0.4643 & 0.7828 & 0.5000 & 0.9898 \\ 0.8775 & 0.9422 & 0.7887 & 0.8775 & 0.7828 \end{bmatrix}.$$

Step 3: Calculate the weights of the DMs through the Eqs. (12)–(17):

$$\lambda_1 = 0.3286, \lambda_2 = 0.3338, \lambda_3 = 0.3376.$$

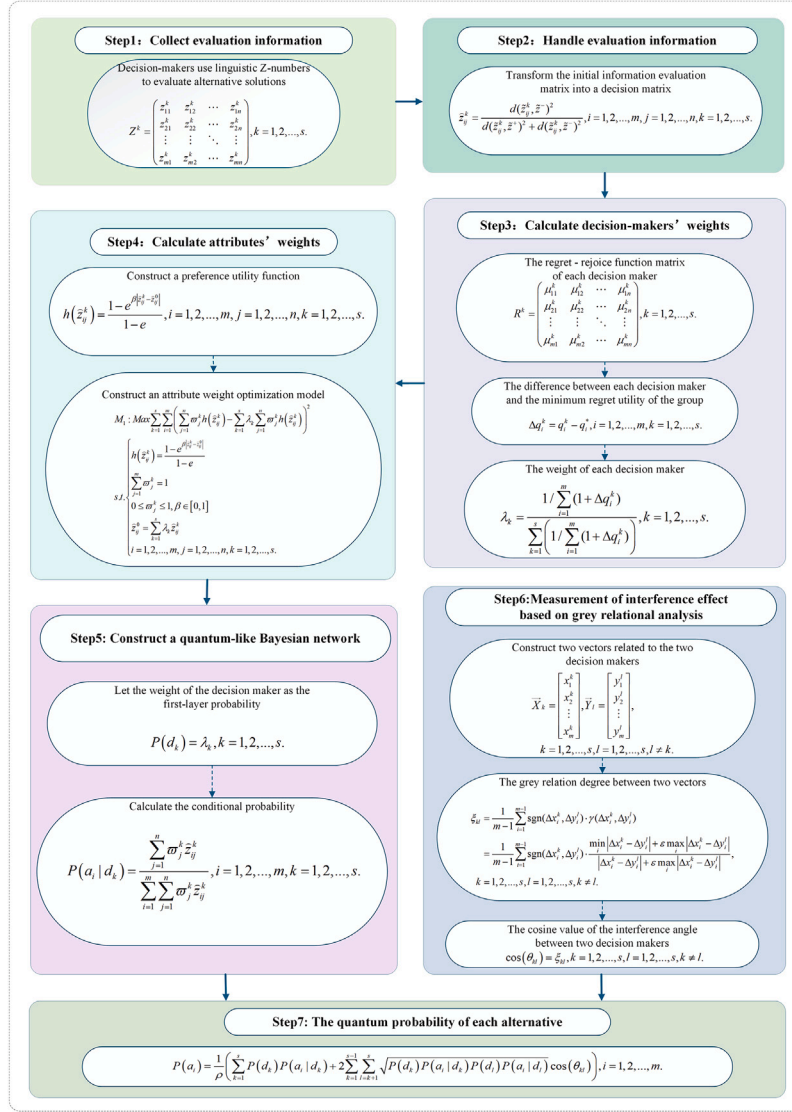


Fig. 2. Decision steps of a MAQGDM model.

Step 4: Calculate the weights of attributes that take into account the preference of the DM by Model (19):

$$W^1 = \{0.2092, 0.3269, 0.2615, 0.0422, 0.1602\},$$

$$W^2 = \{0.0432, 0.2670, 0.3523, 0.2275, 0.1100\},$$

$$W^3 = \{0.0296, 0.3933, 0.0071, 0.5525, 0.0175\}.$$

Step 5: Firstly, set the DMs' weights as the first layer probability of the QLBN:

$$P(d_1) = 0.3286, P(d_2) = 0.3338, P(d_3) = 0.3376.$$

Secondly, the conditional probability of QLBN is calculated according to Eq. (21):

$$P(a_1 | d_1) = 0.2316, P(a_1 | d_2) = 0.1842, P(a_1 | d_3) = 0.2236,$$

$$P(a_2 | d_1) = 0.2587, P(a_2 | d_2) = 0.2164, P(a_2 | d_3) = 0.2168,$$

$$P(a_3 | d_1) = 0.2097, P(a_3 | d_2) = 0.2043, P(a_3 | d_3) = 0.2083,$$

$$P(a_4 | d_1) = 0.1712, P(a_4 | d_2) = 0.2196, P(a_4 | d_3) = 0.1268,$$

$$P(a_5 | d_1) = 0.1288, P(a_5 | d_2) = 0.1755, P(a_5 | d_3) = 0.2245.$$

Step 6: Calculate the cosine of the interference angle by the Eqs. (22)–(28):

$$\cos(\theta_{12}) = 0.0088, \cos(\theta_{13}) = 0.0815, \cos(\theta_{23}) = -0.4428.$$

Step 7: Calculate the quantum probability of each alternative through Eq. (29):

$$P(a_1) = 0.2175, P(a_2) = 0.2357, P(a_3) = 0.2074,$$

$$P(a_4) = 0.1724, P(a_5) = 0.1670.$$

To sum up, the ranking result of each alternative is: $a_2 > a_1 > a_3 > a_4 > a_5$, and the optimal alternative is a_2 .

4.3. Sensitivity analysis

(1) The influence of sensitivity coefficient β change on DMs' weights of attributes

When assigning attribute's weights in MAGDM, sensitivity coefficient β ($\beta \in [0, 1]$) has a significant impact on simulating the preference behaviour of DMs. It controls the reaction rate at which DMs respond to changes in the difference between evaluation values and the group evaluation benchmark, and thus affects the allocation of attribute's

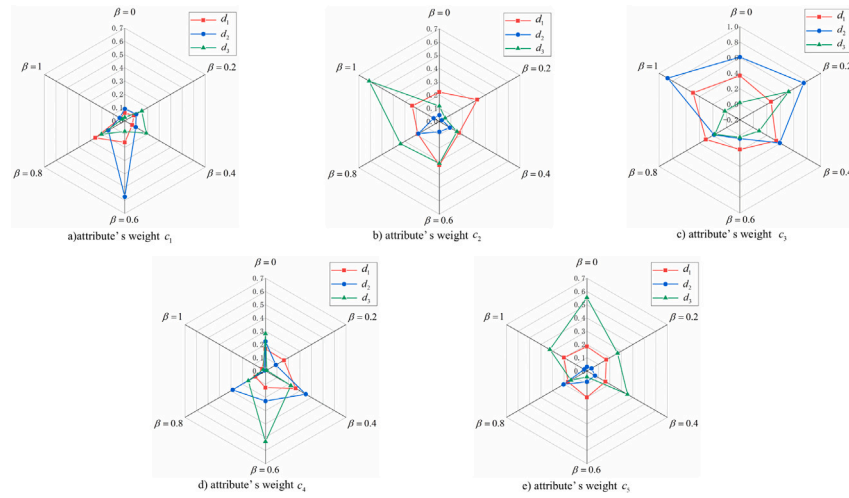


Fig. 3. The weight of each attribute of each DM changes with β .

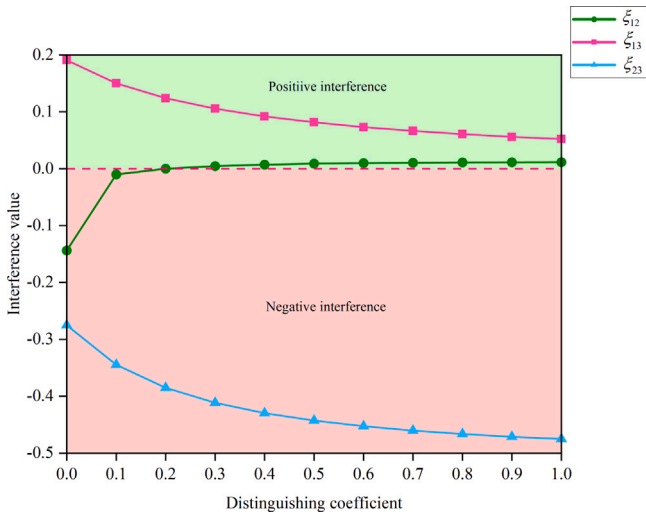


Fig. 4. Variation of interference value with distinguishing coefficient.

weights. Observe the influence of the change of sensitivity coefficient on the DMs' attribute's weights, as shown in Fig. 3:

Fig. 3 shows the influence of the change of sensitivity coefficient β on the attribute's weights of each DM. Firstly, the weight distribution of five attributes of DM d_1 is observed. For attribute c_1 , it shows an obvious increasing trend with the increase of β , and the change is relatively smooth when the value of β is low. The weight of attribute c_2 , attribute c_3 and attribute c_5 show low sensitivity to the change of β , and the weight of attribute c_4 fluctuates greatly under the change of β , indicating that the weight of this attribute is more sensitive to the change of β . Secondly, for DM d_2 , each attribute is sensitive to the change of β . Among them, attribute c_1 , attribute c_2 and attribute c_5 show a large fluctuation under a specific value range of β , and the weight of attribute c_3 has the largest fluctuation relative to the change of β , which shows that this attribute significantly influences DM d_2 's preference adjustment process. Finally, each attribute of DM d_3 shows greater fluctuation with the change of β value compared with DM d_1 and DM d_2 .

All in all, the performance of different attributes under the change of sensitivity coefficient is significantly different. Specifically, when faced with different sensitivity coefficients, DMs are more sensitive to the difference between their own evaluation value and the group

benchmark value under certain attributes, which may lead to increased volatility of decision-making results, thus affecting the robustness and consistency of MAGDM. Therefore, fully considering the sensitivity of DMs to the difference between their own evaluation value and the group benchmark value can help them allocate attributes' weights more effectively in the uncertain environment.

In this paper, $\beta = 0.5$ is selected because for the weight distribution of DMs' attributes, some weights change sharply when β value is small, while others fluctuate greatly when β value is large, and $\beta = 0.5$ can balance these two different trends to a certain extent. At the same time, compared with β taking other values (such as near 0 or 1), when $\beta = 0.5$, the fluctuation of the attribute's weights of each DM is relatively gentle, and DMs will not be too sensitive or insensitive to the change of the difference between the evaluation value and the group evaluation benchmark value, so that the adjustment of attribute's weights can be in a relatively stable and predictable rhythm. It is helpful to form a relatively stable weight allocation scheme in MAGDM.

(2) The influence of resolution coefficient ϵ on the interference intensity

The value of the resolution coefficient ϵ can adjust the sensitivity of the relational coefficient to the difference in extreme values, thereby further influencing the decision-making. In order to explore the influence of resolution coefficient ϵ on interference intensity, this paper conducted a sensitivity analysis, as illustrated in Fig. 4:

Fig. 4 shows the change in interference intensity between the two DMs as the resolution coefficient ϵ increases from 0 to 1. From the numerical performance of different DM combinations, it can be seen that the response of different interference terms to the resolution coefficient varies. For example, ξ_{12} increases as ϵ increases, with a larger slope in the smaller ϵ range [0, 0.5], being more sensitive to parameter changes. After ϵ continues to increase, the slope gradually decreases. Moreover, the interference between DM d_1 and DM d_2 gradually changes from negative interference to positive interference. The interference intensity shows a trend of first decreasing, then increasing, and finally stabilizing as ϵ increases. ξ_{13} decreases as the resolution coefficient ϵ increases, with a larger slope in the smaller ϵ range. ξ_{13} shows a significant decrease in amplitude and then a slow downward trend and gradually stabilizes. The interference values between DM d_1 and DM d_3 are always positive, and the assessment tendencies of these two DMs are generally mutually reinforcing, but the degree of reinforcement decreases as the resolution coefficient increases. The trend of ξ_{23} 's change with the increase of the resolution coefficient ϵ is similar to that of ξ_{13} . DM d_2 and DM d_3 always have negative interference between them, and the interference intensity shows a significant growth trend as ϵ increases. As ϵ increases, the divergence in the assessment of the two

Table 2

Comparison between the proposed model and existing decision models.

Models	Information	DMs' weights	Attributes' weights	DMs' preference	Interference effect	Interference angle measurement method
Jin et al. (2024)	HFLTS	Unknown	Known	Yes	No	No
Gurmani et al. (2024)	LT-SFS	Known	Known	No	No	No
Wang et al. (2014)	PLTS	Unknown	Unknown	No	No	No
Liu, Liao, et al. (2023)	Z-numbers	Unknown	Unknown	No	No	No
She et al. (2021b)	Crisp number	No	Unknown	No	Yes	Deng entropy
Wang, Liu, et al. (2024)	Crisp number	Unknown	Known	No	Yes	Specific phase angle
Xiao et al. (2025)	LDAS	Unknown	No	No	Yes	entropy
Zheng and Liu (2025)	2-LV	Known	No	No	Yes	Heuristic algorithm
Liu and Zhu (2022)	LTS	Unknown	Unknown	No	Yes	Heuristic algorithm
Tan et al. (2024)	LVIFS	Unknown	Unknown	No	Yes	Heuristic algorithm
The proposed model	LZN	Unknown	Unknown	Yes	Yes	Grey relation degree

Note: HFLTS is the hesitant fuzzy linguistic term set, LT-SFS is the linguistic T spherical fuzzy set, PLTS is the probabilistic linguistic term set, LZN is the linguistic Z-numbers, LDAS is the linguistic distribution evaluation, 2-LV is the binary linguistic number, LTS is the linguistic term set, and LVIFS is the linguistic value intuitionistic fuzzy set.

DMs is greater, and the situation with a larger difference in alternatives is given a higher weight. From the overall trend, the intensity of the interference effect between the two DMs changes significantly in the smaller range of the resolution coefficient ϵ , indicating that their interference effect is relatively significant. As the resolution coefficient ϵ continues to increase, the intensity of the interference effect gradually stabilizes. ϵ is the weight of the maximum term in the grey relational degree calculation formula. When ϵ is small, the grey relational coefficient is more sensitive to the local minimum values in the evaluation difference, and the interference term is prone to large fluctuations, thus presenting a more obvious increasing or decreasing trend. As ϵ increases, the maximum term in the evaluation difference gradually occupies a dominant position in the relational coefficient calculation, thus making the numerical distribution of the interference term stable. This indicates that the constructed interference effect measurement method has good numerical stability and robustness within a wide parameter range. The difference in the response of different DM combinations to the amplitude of the interference term reflects the flexibility of the model in depicting the structure of evaluation differences. By analysing the regulatory mechanism and dynamic differences of the interference effect between DMs in group decision-making controlled by the resolution coefficient, it provides important practical support for improving the stability and interpretability of the decision model.

4.4. Comparative analysis

4.4.1. Qualitative comparative analysis

To clearly distinguish the proposed approach from existing models, a comparative analysis of existing typical studies was conducted (as shown in Table 2), aiming to illustrate the uniqueness of the proposed method in this paper.

It can be obtained from Table 2 that traditional group decision-making methods typically rely on the assumption of DM independence, neither accounting for interference effect among DMs nor adequately addressing psychological characteristic modelling in complex uncertain environments. Furthermore, most of the existing methods for measuring the interference effect of quantum decision-making models are sensitive to the data distribution or directly assume it as a specific phase Angle for evolutionary analysis, resulting in low accuracy of decision-making results. Moreover, they only focus on the modelling of the interference effect, thereby simplifying the objective computation of the weights of DMs and attributes, and lacking practical guidance.

Unlike existing methods, the proposed approach simultaneously considers interference effects among DMs and accounts for their psychological preference differences and other factors caused by cognitive differences in complex uncertain environments. Subsequently, the optimal attribute's weights are derived through constructing an optimization model. Furthermore, this paper measures the interference

effect based on the grey relational degree. It can effectively deal with uneven data distribution and handle nonlinear relationships effectively. Thus, the proposed method in this study demonstrates stronger applicability in uncertain decision-making environments, along with more prominent robustness and stability performance.

4.4.2. Quantitative comparative analysis

To verify the effectiveness of the method in this paper, three classical group decision-making methods and three quantum group decision-making methods are selected for comparison. These methods were applied to the data set in this paper and the priority order of the alternatives was compared and analysed. Among them, MAGDM Model 1 (Li et al., 2022) represents the TOPSIS method, MAGDM Model 2 (Sun et al., 2023) represents the TODIM method, and MAGDM Model 3 (Li et al., 2025) represents the PROMETHEE method. The comparison results are shown in Table 3.

It can be seen from the comparison results that compared with other models, the same results can be selected by the proposed model, which proves the effectiveness and applicability of the proposed model, but there are differences in the ordering of specific alternatives. Compared with the traditional model, the optimal alternative is a_2 , but the ranking results of other alternatives differ significantly from the results in this paper. For example, MAGDM model 1 (Li et al., 2022) calculates the distance and similarity between the alternatives and the positive and negative ideal solutions, and finally sorts the alternatives according to the similarity and selects the best solution. MAGDM model 2 (Sun et al., 2023) completes the ranking of alternatives by calculating their overall dominance degrees. MAGDM model 3 (Li et al., 2025) constructs a comprehensive preference based on the preference function using the PROMETHEE method, and then calculates the positive and negative flows to complete the alternative ranking. However, none of these three methods take into account the interference effect among DMs, which makes it difficult to accurately reflect the interaction of opinions in complex decision-making scenarios, thereby affecting the ranking results of other alternatives.

Compared with the quantum model, the optimal alternative is also a_2 , and the ranking results of other alternatives are also quite different from those of this paper. Quantum model 1 (Wu et al., 2021) uses the generalized TODIM method to calculate the global dominance of various alternatives and then solve the conditional probability. However, it assumes predetermined weights for both attributes and DMs and does not involve the calculation of interference angles. Consequently, the model exhibits marked deterioration in both decision-making precision and stability when handling complex tasks. Quantum model 2 (She et al., 2021a) determines the cosine value of the interference angle by constructing a connection vector and calculating Deng entropy to measure the degree of interference between attributes. In contrast, the method of characterizing the interference angle based on Deng entropy

Table 3
Comparison of ranking results among different methods.

Method	Ranking indices	Final ranking orders
MAGDM Model 1 (Li et al., 2022)	$S_1 = 0.5328, S_2 = 0.6743, S_3 = 0.3427,$ $S_4 = 0.4254, S_5 = 0.3504$	$a_2 > a_1 > a_4 > a_5 > a_3$
MAGDM Model 2 (Sun et al., 2023)	$\varphi(a_1) = -1.1616, \varphi(a_2) = 1.3873, \varphi(a_3) = -2.3315,$ $\varphi(a_4) = -2.0226, \varphi(a_5) = -2.5301$	$a_2 > a_1 > a_4 > a_3 > a_5$
MAGDM Model 3 (Li et al., 2025)	$\Phi(a_1) = 0.4291, \Phi(a_2) = 0.6283, \Phi(a_3) = -0.3556,$ $\Phi(a_4) = -0.3486, \Phi(a_5) = -1.3371$	$a_2 > a_1 > a_4 > a_3 > a_5$
Quantum model 1 (Wu et al., 2021)	$P(a_1) = 0.1530, P(a_2) = 0.5488, P(a_3) = 0.0994,$ $P(a_4) = 0.1125, P(a_5) = 0.0863$	$a_2 > a_1 > a_4 > a_3 > a_5$
Quantum model 2 (She et al., 2021a)	$P(a_1) = 0.2041, P(a_2) = 0.2513, P(a_3) = 0.1761,$ $P(a_4) = 0.1848, P(a_5) = 0.1836$	$a_2 > a_1 > a_4 > a_5 > a_3$
Quantum model 3 (Wan et al., 2026)	$P(a_1) = 0.2078, P(a_2) = 0.2332, P(a_3) = 0.1871,$ $P(a_4) = 0.1918, P(a_5) = 0.1801$	$a_2 > a_1 > a_4 > a_3 > a_5$
The proposed model	$P(a_1) = 0.2175, P(a_2) = 0.2357, P(a_3) = 0.2074,$ $P(a_4) = 0.1724, P(a_5) = 0.1670$	$a_2 > a_1 > a_3 > a_4 > a_5$

focuses more on reflecting the group evaluation structure from the perspective of overall uncertainty and information dispersion. However, this method is sensitive to the sample size and evidence allocation structure. In small samples or highly sparse information situations, its characterization of the specific interference relationship among DMs may have certain instability. Quantum model 3 (Wan et al., 2026) measures the interference angle using a similarity heuristic algorithm, but it also does not take into account the psychological preferences of DMs and nonlinear psychological reactions. This paper measures the interference effect between DMs using the grey relational degree and considers the psychological sensitivity and cognitive response characteristics of DMs to changes in different attribute's weights, optimizing the determination of attribute's weights. Therefore, there are certain differences in the ranking results of other alternatives between this paper and these three methods.

5. Conclusion

To address the challenge of decision-making in complex and uncertain environments within the industrial manufacturing sector, this paper proposes a MAQGD model considering the preference of DMs. Firstly, based on regret theory, this paper evaluates the relative regret degree felt by DMs under different decision choices, and then calculates the weights of DMs. In addition, a preference utility function is proposed to reflect the change of DMs' preferences and quantify the importance of different attributes. On this basis, the optimal attribute's weights of each DM are derived through constructing an optimization model. Meanwhile, a QLBN is established to better handle the complex interference effects among DMs, and the positive and negative grey relational degrees are expanded to describe the direction and intensity of the mutual influence between DMs, thereby measuring the interference effects among DMs and calculating the quantum probability of each alternative. The results show that this model can quantify the psychological preferences and relative regret of DMs, and utilize the quantum probability theory to depict the nonlinear interference among DMs, providing more scientific support for decision-making. Moreover, in MAGDM, fully considering the sensitivity of DMs to the differences between their own evaluation values and the group benchmark values can help them more effectively allocate attribute's weights in an uncertain environment.

This paper not only deepens the application of quantum decision theory in the field of group decision-making, but also has important guiding value for industrial manufacturing practices. Although the experimental verification of this study focuses on the logistics equipment selection scenario, the proposed decision model has good theoretical universality and method transferability. For example: (1) In supplier selection, joint inventory management, or sustainable supply chain design, participating enterprises often have conflicts of interest and psychological expectation differences. This model can capture the

psychological avoidance tendencies of all parties when weighing economic, environmental and social attributes through the regret theory, and analyse the interactive influence of negotiations and compromises between enterprises through quantum interference, thereby supporting more robust collaborative decision-making. (2) In the field of energy planning, DMs often have significant differences in preferences for different planning schemes, and there is a significant interference effect between viewpoints. This model can quantify the psychological preferences and relative regret degrees of DMs, and use quantum probability theory to depict the nonlinear interference between the viewpoints of different scheme supporters, providing more scientific support for decision-making.

Although this paper expands the research on multi-attribute quantum group decision-making, it still has some limitations. Firstly, the construction and solution process of quantum probability model involves complex mathematical operations, resulting in large computational resource requirements, and the computational complexity will increase significantly with the cardinality of both decision attributes and DMs. Secondly, the consideration of the intrinsic correlation between attributes in this paper is insufficient.

Further research could explore the following research directions in the future. Firstly, for the high computational complexity of quantum probability model and utility preference function modelling, more efficient algorithms or approximate computational methods can be explored to enhance computational efficiency and model applicability. Secondly, complex network analysis methods can be applied to construct an attribute relationship network, further exploring the correlation between attributes, while considering the impact of decision-maker heterogeneity on attribute correlation, to provide more accurate basis for MAGDM.

CRedit authorship contribution statement

Shuli Yan: Writing – review & editing, Methodology, Conceptualization. **Ruyi Bian:** Writing – original draft, Methodology. **Yizhao Xu:** Writing – review & editing, Visualization. **Feixia Ji:** Writing – review & editing, Methodology. **Francisco Javier Cabrerizo:** Writing – review & editing, Validation, Supervision.

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Data availability

No data was used for the research described in the article.

References

- Bell, D. E. (1982). Regret in decision making under uncertainty. *Operations Research*, 30(5), 961–981. <http://dx.doi.org/10.1287/opre.30.5.961>.
- Cao, M., Jin, Y., Hu, Y., Huang, X., & Wu, J. (2024). A decision framework for Chinese-style cruise ship design based on informativeness weight method and group consensus reaching model. *Advanced Engineering Informatics*, 62, Article 102698. <http://dx.doi.org/10.1016/j.aei.2024.102698>.
- Chai, J., Su, Y., & Lu, S. (2023). Linguistic Z-number preference relation for group decision making and its application in digital transformation assessment of SMEs. *Expert Systems with Applications*, 213, Article 118749. <http://dx.doi.org/10.1016/j.eswa.2022.118749>.
- Chen, B., Cai, Q., Wei, G., & Mo, Z. (2023). Novel aczel-alsina operations-based linguistic Z-number aggregation operators and their applications in multi-attribute group decision-making process. *Engineering Applications of Artificial Intelligence*, 124, Article 106541. <http://dx.doi.org/10.1016/j.engappai.2023.106541>.
- Dong, J., & Wan, S. (2024). Interval-valued intuitionistic fuzzy best-worst method with additive consistency. *Expert Systems with Applications*, 236, Article 121213. <http://dx.doi.org/10.1016/j.eswa.2023.121213>.
- Du, Z., Yu, S., Guo, L., & Zhong, X. (2024). Multi-stage type- α constrained minimum-cost consensus for linguistic Z-number large-scale decision-making. *Engineering Applications of Artificial Intelligence*, 136, Article 109032. <http://dx.doi.org/10.1016/j.engappai.2024.109032>.
- Fan, S., Liang, H., Li, C.-C., Chiclana, F., Pedrycz, W., & Dong, Y. (2024). Optimal resources allocation to support the consensus reaching in group decision making. *Information Fusion*, 110, Article 102451. <http://dx.doi.org/10.1016/j.inffus.2024.102451>.
- Farrokhzadeh, E., Seyfi-Shishavan, S. A., Kutlu Gündoğdu, F., Donyatalab, Y., Kahraman, C., & Seifi, S. H. (2021). A spherical fuzzy methodology integrating maximizing deviation and TOPSIS methods. *Engineering Applications of Artificial Intelligence*, 101, Article 104212. <http://dx.doi.org/10.1016/j.engappai.2021.104212>.
- Gurmani, S. H., Zhang, S., Awwad, F. A., & Ismail, E. A. (2024). Combinative distance-based assessment method using linguistic T-spherical fuzzy aggregation operators and its application to multi-attribute group decision-making. *Engineering Applications of Artificial Intelligence*, 133, Article 108165. <http://dx.doi.org/10.1016/j.engappai.2024.108165>.
- Ji, F., Wu, J., Chiclana, F., Sun, Q., & Herrera-Viedma, E. (2024). A trust incentive driven feedback mechanism with risk attitude for group consensus in social networks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 1–14. <http://dx.doi.org/10.1109/TSMC.2024.3519510>.
- Ji, F., Wu, J., Chiclana, F., Sun, Q., Liang, C., & Herrera-Viedma, E. (2024). Decayed trust propagation method in multiple overlapping communities for improving consensus under social network group decision making. *IEEE Transactions on Fuzzy Systems*, 32(8), 4401–4412. <http://dx.doi.org/10.1109/TFUZZ.2024.3398591>.
- Jiang, J., Liu, X., Garg, H., & Zhang, S. (2023). Large group decision-making based on interval rough integrated cloud model. *Advanced Engineering Informatics*, 56, Article 101964. <http://dx.doi.org/10.1016/j.aei.2023.101964>.
- Jiang, J., Liu, X., Wang, Z., Ding, W., Zhang, S., & Xu, H. (2024). Large group decision-making with a rough integrated asymmetric cloud model under multi-granularity linguistic environment. *Information Sciences*, 678, Article 120994. <http://dx.doi.org/10.1016/j.ins.2024.120994>.
- Jin, F., Lin, H., & Zhou, L. (2024). Bounded rationality consensus reaching process with prospect theory and preventing individual weight manipulation for multi-attribute group decision making. *Information Fusion*, 112, Article 102574. <http://dx.doi.org/10.1016/j.inffus.2024.102574>.
- Kumar, R., & Kumar, S. (2024). An extended combined compromise solution framework based on novel intuitionistic fuzzy distance measure and score function with applications in sustainable biomass crop selection. *Expert Systems with Applications*, 239, Article 122345. <http://dx.doi.org/10.1016/j.eswa.2023.122345>.
- Li, D. (2024). A linguistic Z-number-based dual perspectives information volume calculation method for driving behavior risk evaluation. *Expert Systems with Applications*, 257, Article 124992. <http://dx.doi.org/10.1016/j.eswa.2024.124992>.
- Li, G., Kou, G., & Peng, Y. (2022). Heterogeneous large-scale group decision making using fuzzy cluster analysis and its application to emergency response plan selection. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 52(6), 3391–3403. <http://dx.doi.org/10.1109/TSMC.2021.3068759>.
- Li, J., Xiang, Y., Peng, H., & Wang, J. (2025). An extended multi-criteria group decision-making method based on preference ranking under Z-number environments. *Engineering Applications of Artificial Intelligence*, 149, Article 110573. <http://dx.doi.org/10.1016/j.engappai.2025.110573>.
- Liao, H., Wu, Z., Tang, M., & Wan, Z. (2022). An interactive consensus reaching model with updated weights of clusters in large-scale group decision making. *Engineering Applications of Artificial Intelligence*, 107, Article 104532. <http://dx.doi.org/10.1016/j.engappai.2021.104532>.
- Liu, S. (2023). Negative grey relational model and measurement of the reverse incentive effect of fields medal. *Grey Systems: Theory and Application*, 13(1), 1–13. <http://dx.doi.org/10.1108/GS-10-2021-0148>.
- Liu, S., & Forrest, J. Y. L. (2010). *Grey systems: Theory and applications*. Springer.
- Liu, F., Liao, H., Wu, X., & Al-Barakati, A. (2023). Evaluating internet hospitals by a linguistic Z-number-based gained and lost dominance score method considering different risk preferences of experts. *Information Sciences*, 630, 647–668. <http://dx.doi.org/10.1016/j.ins.2023.02.061>.
- Liu, Y., Liu, X., Jiang, J., & Han, S. (2023). A trust-based quantum probabilistic linguistic multi-criteria group decision making model considering interference effect. *Computers & Industrial Engineering*, 184, Article 109503. <http://dx.doi.org/10.1016/j.cie.2023.109503>.
- Liu, J., Shao, L., Jin, F., & Tao, Z. (2024). A multi-attribute group decision-making method based on trust relationship and DEA regret cross-efficiency. *IEEE Transactions on Engineering Management*, 71, 824–836. <http://dx.doi.org/10.1109/TEM.2021.3138970>.
- Liu, N., Xu, Z., & Wu, H. (2023). Decision field theory-combined multi-attribute group decision-making method for incomplete linear ordinal ranking. *Applied Soft Computing*, 140, Article 110056. <http://dx.doi.org/10.1016/j.asoc.2023.110056>.
- Liu, W., & Zhu, J. (2022). A multistage decision-making method with quantum-guided expert state transition based on normal cloud models. *Information Sciences*, 615, 700–730. <http://dx.doi.org/10.1016/j.ins.2022.10.056>.
- Loomes, G., & Sugden, R. (1982). Regret theory: An alternative theory of rational choice under uncertainty. *The Economic Journal*, 92(368), 805–824. <http://dx.doi.org/10.2307/2232669>.
- Ma, X., Han, X., Xu, Z., Rodríguez, R. M., & Zhan, J. (2025). Fusion of probabilistic linguistic term sets for enhanced group decision-making: Foundations, survey and challenges. *Information Fusion*, 116, Article 102802. <http://dx.doi.org/10.1016/j.inffus.2024.102802>.
- Meşe Erdoğan, E., Yılmaz, H., Topuz, E., & Özkan, M. (2024). Decision making with analytical hierarchy process approach to select feedstock type for bioethanol production: Is 3G more sustainable than others? *Renewable Energy*, 236, Article 121464. <http://dx.doi.org/10.1016/j.renene.2024.121464>.
- Mi, X., Tang, M., Liao, H., Shen, W., & Lev, B. (2019). The state-of-the-art survey on integrations and applications of the best worst method in decision making: Why, what, what for and what's next? *Omega*, 87, 205–225. <http://dx.doi.org/10.1016/j.omega.2019.01.009>.
- R., V., Mahapatra, G., & Pamucar, D. (2024). Fermatean trapezoidal fuzziness average aggregation scheme for selection of infant clothing by group decision-making. *Applied Soft Computing*, 167, Article 112317. <http://dx.doi.org/10.1016/j.asoc.2024.112317>.
- Rani, P., & Mishra, A. R. (2021). Fermatean fuzzy Einstein aggregation operators-based multimooora method for electric vehicle charging station selection. *Expert Systems with Applications*, 182, Article 115267. <http://dx.doi.org/10.1016/j.eswa.2021.115267>.
- Ranjbar, M., Effati, S., & Miri, S. (2022). Fully hesitant fuzzy linear programming with hesitant fuzzy numbers. *Engineering Applications of Artificial Intelligence*, 114, Article 105047. <http://dx.doi.org/10.1016/j.engappai.2022.105047>.
- She, L., Han, S., & Liu, X. (2021a). Application of quantum-like Bayesian network and belief entropy for interference effect in multi-attribute decision making problem. *Computers & Industrial Engineering*, 157, Article 107307. <http://dx.doi.org/10.1016/j.cie.2021.107307>.
- She, L., Han, S., & Liu, X. (2021b). Application of quantum-like Bayesian network and belief entropy for interference effect in multi-attribute decision making problem. *Computers & Industrial Engineering*, 157, Article 107307. <http://dx.doi.org/10.1016/j.cie.2021.107307>.
- Shen, Y., Ma, X., Zhang, H., & Zhan, J. (2024). Fusion social network and regret theory for a consensus model with minority opinions in large-scale group decision making. *Information Fusion*, 112, Article 102548. <http://dx.doi.org/10.1016/j.inffus.2024.102548>.
- Shen, K., Wang, X., Qiao, D., & Wang, J. (2020). Extended Z-MABAC method based on regret theory and directed distance for regional circular economy development program selection with Z-information. *IEEE Transactions on Fuzzy Systems*, 28(8), 1851–1863. <http://dx.doi.org/10.1109/TFUZZ.2019.2923948>.
- Sun, H., Yang, Z., Cai, Q., Wei, G., & Mo, Z. (2023). An extended exp-TODIM method for multiple attribute decision making based on the Z-wasserstein distance. *Expert Systems with Applications*, 214, Article 119114. <http://dx.doi.org/10.1016/j.eswa.2022.119114>.
- Tan, X., Zhu, J., Wei, G., & Tong, H. (2024). The dual interactions-driven consensus model: Probing the non-cooperative behavior through order effects. *Information Fusion*, 105, Article 102212. <http://dx.doi.org/10.1016/j.inffus.2023.102212>.
- Tang, G., Zhang, T., & Liu, P. (2025). Conflict resolution under power asymmetry based on incomplete linguistic preference relations. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 55(12), 9761–9774. <http://dx.doi.org/10.1109/TSMC.2025.3621259>.
- Van Dijk, E., & De Dreu, C. K. (2021). Experimental games and social decision making. *Annual Review of Psychology*, 72(1), 415–438. <http://dx.doi.org/10.1146/annurev-psych-081420-110718>.
- van Hoef, R., Connell, L., & Lynott, D. (2023). The effects of sensorimotor and linguistic information on the basic-level advantage. *Cognition*, 241, Article 105606. <http://dx.doi.org/10.1016/j.cognition.2023.105606>.

- Wan, S., Dong, J., & Chen, S.-M. (2021). Fuzzy best-worst method based on generalized interval-valued trapezoidal fuzzy numbers for multi-criteria decision-making. *Information Sciences*, 573, 493–518. <http://dx.doi.org/10.1016/j.ins.2021.03.038>.
- Wan, S.-P., Dong, J.-Y., & Chen, S.-M. (2024). An integrated method for shared power bank supplier selection based on linguistic hesitant fuzzy multi-criteria group decision making. *Knowledge-Based Systems*, 301, Article 112300. <http://dx.doi.org/10.1016/j.knsys.2024.112300>.
- Wan, S.-P., Yan, J., Dong, J.-Y., & Gao, Y. (2026). A dual-similarity based consensus reaching process in quality function deployment with heterogeneous linguistic preference relations. *Engineering Applications of Artificial Intelligence*, 163, Article 113051. <http://dx.doi.org/10.1016/j.engappai.2025.113051>.
- Wang, F. (2021). Preference degree of triangular fuzzy numbers and its application to multi-attribute group decision making. *Expert Systems with Applications*, 178, Article 114982. <http://dx.doi.org/10.1016/j.eswa.2021.114982>.
- Wang, H., Ju, Y., Dong, P., Wang, A., & Cabrerizo, F. J. (2024). Preference-based regret three-way decision method on multiple decision information systems with linguistic Z-numbers. *Information Sciences*, 654, Article 119861. <http://dx.doi.org/10.1016/j.ins.2023.119861>.
- Wang, P., Liu, P., Li, Y., Teng, F., & Pedrycz, W. (2024). Trust exploration- and leadership incubation- based opinion dynamics model for social network group decision-making: A quantum theory perspective. *European Journal of Operational Research*, 317(1), 156–170. <http://dx.doi.org/10.1016/j.ejor.2024.03.025>.
- Wang, J., Ma, X., Xu, Z., & Zhan, J. (2022). Regret theory-based three-way decision model in hesitant fuzzy environments and its application to medical decision. *IEEE Transactions on Fuzzy Systems*, 30(12), 5361–5375. <http://dx.doi.org/10.1109/TFUZZ.2022.3176686>.
- Wang, S., Pan, X.-H., Martínez, L., & Moreno-Albarracín, A. (2024). A novel interval type-2 fuzzy consensus reaching process model and group decision-making method for renewable energy investment. *Engineering Applications of Artificial Intelligence*, 133, Article 108422. <http://dx.doi.org/10.1016/j.engappai.2024.108422>.
- Wang, F., Pan, Y., Xu, G., & Wei, Q. (2025). Probabilistic linguistic multi-attribute group decision making method considering the important degrees of experts and attributes. *Expert Systems with Applications*, 265, Article 125899. <http://dx.doi.org/10.1016/j.eswa.2024.125899>.
- Wang, J., Wu, J., Wang, J., Zhang, H., & Chen, X. (2014). Interval-valued hesitant fuzzy linguistic sets and their applications in multi-criteria decision-making problems. *Information Sciences*, 288, 55–72. <http://dx.doi.org/10.1016/j.ins.2014.07.034>.
- Wen, Z., Liao, H., & Figueira, J. R. (2024). A preference disaggregation-driven multiple criteria sorting model based on regret theory. *Omega*, 129, Article 103150. <http://dx.doi.org/10.1016/j.omega.2024.103150>.
- Wu, Q., Liu, X., Qin, J., Wang, W., & Zhou, L. (2021). A linguistic distribution behavioral multi-criteria group decision making model integrating extended generalized TODIM and quantum decision theory. *Applied Soft Computing*, 98, Article 106757. <http://dx.doi.org/10.1016/j.asoc.2020.106757>.
- Wu, Q., Liu, X., Zhou, L., Tao, Z., & Qin, J. (2022). A quantum framework for modeling interference effects in linguistic distribution multiple criteria group decision making. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 52(6), 3492–3507. <http://dx.doi.org/10.1109/TSMC.2021.3072388>.
- Xiao, J., Cai, M., Wei, G., & Hu, S. (2025). An adaptive consensus model with hybrid feedback mechanism: Exploring interference effects under evidence theory. *Information Fusion*, 118, Article 102949. <http://dx.doi.org/10.1016/j.inffus.2025.102949>.
- Xiao, Y., Li, M., Li, Y., Xu, Y., & Liu, H. (2026). Green smart home design based on quality function deployment integrating customer requirements and life-cycle sustainability in Z-numbers environment. *Engineering Applications of Artificial Intelligence*, 164, Article 112901. <http://dx.doi.org/10.1016/j.engappai.2025.112901>.
- Xing, Y., Wu, J., Chiclana, F., Wang, S., & Zhu, Z. (2025). Personalized trust incentive mechanisms with personality characteristics for minimum cost consensus in group decision making. *Information Fusion*, 118, Article 102967. <http://dx.doi.org/10.1016/j.inffus.2025.102967>.
- Xu, Z., Xu, H., Li, P., & Wei, C. (2024). Social network group decision-making method based on stochastic multi-criteria acceptability analysis for probabilistic linguistic term sets. *Information Sciences*, 681, Article 121269. <http://dx.doi.org/10.1016/j.ins.2024.121269>.
- Xu, Y., Yan, S., & Li, Y. (2025). A multi-attribute quantum group consensus model considering psychological preference. *Engineering Applications of Artificial Intelligence*, 144, Article 110086. <http://dx.doi.org/10.1016/j.engappai.2025.110086>.
- Xu, Y., Yan, S., Pérez, I. J., & Cabrerizo, F. J. (2025). Quantum group consensus model based on linguistic Z-numbers and its application to meteorological disaster emergency. *Information Sciences*, 700, Article 121777. <http://dx.doi.org/10.1016/j.ins.2024.121777>.
- Yan, J., Dong, J.-Y., Wan, S.-P., & Gao, Y. (2025). A quantum probability theory-based method for heterogeneous multi-criteria group decision making with incomplete probabilistic linguistic preference relations considering the interference effect among decision makers. *Journal of the Operational Research Society*, 76(6), 1225–1251. <http://dx.doi.org/10.1080/01605682.2024.2417726>.
- Yan, S., Xu, Y., Gong, Z., & Herrera-Viedma, E. (2024). A quantum group decision model for meteorological disaster emergency response based on D-S evidence theory and choquet integral. *Information Sciences*, 674, Article 120707. <http://dx.doi.org/10.1016/j.ins.2024.120707>.
- Yan, S., Xu, Y., Gong, Z., & Herrera-Viedma, E. (2025). Fuzzy quantum group decision making and its application in meteorological disaster emergency. *IEEE Transactions on Fuzzy Systems*, 1–13. <http://dx.doi.org/10.1109/TFUZZ.2024.3525009>.
- Yan, S., Zeng, Y., Gong, Z., & Wang, J. (2024). Multi-attribute quantum group interference decision model considering minimum constraint and information entropy. *Computers & Industrial Engineering*, 192, Article 110255. <http://dx.doi.org/10.1016/j.cie.2024.110255>.
- Yang, X., Xu, Z., & Xu, J. (2023). Large-scale group delphi method with heterogeneous decision information and dynamic weights. *Expert Systems with Applications*, 213, Article 118782. <http://dx.doi.org/10.1016/j.eswa.2022.118782>.
- Zadeh, L. A. (2011). A note on Z-numbers. *Information Sciences*, 181(14), 2923–2932. <http://dx.doi.org/10.1016/j.ins.2011.02.022>.
- Zhang, N., Zhou, Y., Liu, J., & Wei, G. (2023). VIKOR method for pythagorean hesitant fuzzy multi-attribute decision-making based on regret theory. *Engineering Applications of Artificial Intelligence*, 126, Article 106857. <http://dx.doi.org/10.1016/j.engappai.2023.106857>.
- Zhang, S., Zhu, J., & Tong, H. (2024). Risk assessment for complex product delivery: Group decision-making incorporating personalized cognition and dual belief superposition. *Engineering Applications of Artificial Intelligence*, 133, Article 108372. <http://dx.doi.org/10.1016/j.engappai.2024.108372>.
- Zhao, T., Li, Z., & Deng, Y. (2024). Linearity in deng entropy. *Chaos, Solitons & Fractals*, 178, Article 114388. <http://dx.doi.org/10.1016/j.chaos.2023.114388>.
- Zhao, C., Xu, X., Liu, R., & He, J. (2023). A novel multi-attribute group decision-making method for heterogeneous preferences and attribute interaction contexts. *Expert Systems with Applications*, 221, Article 119752. <http://dx.doi.org/10.1016/j.eswa.2023.119752>.
- Zheng, Q., & Liu, X. (2025). A consensus-based multi-criteria decision making method integrating GLDS method and quantum probability theory for risk analysis of human errors. *Computers & Industrial Engineering*, 200, Article 110847. <http://dx.doi.org/10.1016/j.cie.2024.110847>.
- Zhu, Z., Zhang, X., Cao, M., Chiclana, F., & Wu, J. (2025). A dynamic trust and prospect theory driven bilateral feedback mechanism for maximizing consensus income in social network group decision making. *Information Fusion*, 117, Article 102904. <http://dx.doi.org/10.1016/j.inffus.2024.102904>.